

Perfect 3D Lamp Printing Guide

Settings, Profiles & Pro Tips

3D LAMP PACK™

FDM Settings for Lamps

Recommended Print Settings

SETTING	STANDARD MODE	VASE MODE	NOTES
Layer Height	0.2 mm	0.2–0.28 mm	Lower = smoother surface, longer print
Nozzle Temp	200–210 °C	200–210 °C	PLA; adjust per filament brand
Bed Temp	60 °C	60 °C	Use glue stick on glass beds
Print Speed	40–60 mm/s	30–50 mm/s	Slower = better quality for display
Infill	5–10%	N/A (0%)	Lamps don't need structural infill
Wall Count	2–3	1 (single wall)	Thinner walls = better light diffusion
Top Layers	3–4	0	Vase mode lamps are open at the top

Vase Mode (Spiralize Outer Contour)

Vase mode is the secret weapon for lamp printing. The printer creates a single continuous spiral wall with no seams, no infill, and no retractions. This produces lamps that are thin enough for beautiful light diffusion with zero layer-line visibility at the seam.

In your slicer, find it under:

- **Cura:** Special Modes → **Spiralize Outer Contour**
- **PrusaSlicer / OrcaSlicer:** Print Settings → Layers → **Spiral vase**

Pro Tip: Before slicing in vase mode, preview the toolpath. Ensure the model has no internal geometry that would break the spiral. If the lamp has a separate base, print the base in standard mode and the shade in vase mode.

Support Settings for Overhangs

Most lamp designs in this pack are optimized to print without supports. For models with significant overhangs (60°+), use tree supports in your slicer — they are easier to remove and leave a cleaner surface. Set support density to 5–10% and Z-distance to 0.2 mm.

Recommended Slicers

SLICER	BEST FOR	PRICE
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Cura	Beginners, widest printer support	Free
PrusaSlicer	Advanced settings, great defaults	Free
OrcaSlicer	Bambu Lab printers, speed profiles	Free

Resin Settings & Finishing

When to Use Resin vs. FDM for Lamps

FDM is the go-to for most lamp shades because it handles large, hollow prints efficiently and produces great light diffusion with thin walls. Resin printing excels for small, highly detailed lamp components: intricate bases, decorative caps, or small accent lights where crisp detail matters more than size.

FACTOR	FDM	RESIN (SLA/MSLA)
Print size	Large (20–30+ cm)	Small–medium (up to 15 cm)
Detail level	Good	Excellent
Light diffusion	Excellent with PLA	Limited (most resin is opaque)
Cost per lamp	\$0.50–\$2	\$3–\$8
Print time	3–8 hours	4–12 hours + curing

Post-Processing for a Premium Finish

Sanding & Smoothing

For lamps you want opaque or painted, sand with 200-grit to remove layer lines, then 400-grit for smoothness. For display-quality finishes, go up to 800-grit. Wet sanding with 600+ grit produces a near-glass surface on PLA.

Priming & Painting

Apply filler primer (Rust-Oleum or similar) in thin coats. Sand between coats. Finish with spray paint in your desired color. For metallic lamp bases, use metallic spray paint followed by a clear coat.

LED Diffusion Tips

- **Translucent PLA** (natural, white, or pastel) gives the best glow.
- Print with **1–2 walls** (0.4–0.8 mm thick) for maximum light transmission.
- Use **warm white LEDs** (2700–3000K) for a cozy, premium look.
- Place the LED **below the thinnest section** for even diffusion; avoid hotspots by diffusing through frosted acrylic if needed.

Common Problems & Fixes

PROBLEM	LIKELY CAUSE	FIX
Visible layer lines	Layer height too high	Lower to 0.12–0.16 mm, reduce speed
Lamp not diffusing light	Walls too thick or dark filament	Use 1 wall, white/natural PLA, vase mode
Warping at base	Poor bed adhesion	Clean bed, use glue stick, 8-line brim
Stringing inside shade	Retraction settings	Enable retraction 5mm at 45mm/s, use vase mode
Seam line visible	Z-seam alignment	Set Z-seam to "sharpest corner" or use vase mode
Print failing mid-way	Overheating or clog	Clean nozzle, check PTFE tube, improve cooling

Safety Note: Always use LED bulbs (not incandescent) with 3D printed lamps. PLA softens at ~60 °C. LED bulbs produce minimal heat and are safe for PLA enclosures. Never exceed 5W LED inside a fully enclosed PLA shade.

Next Step → Choose your first 3 designs from the Premium Pack (Bonus #3) and print them in vase mode with white PLA.